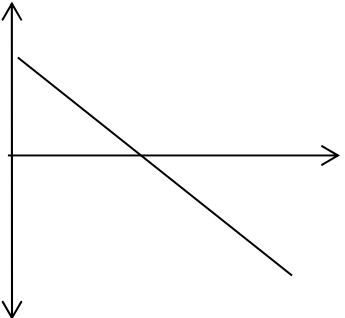
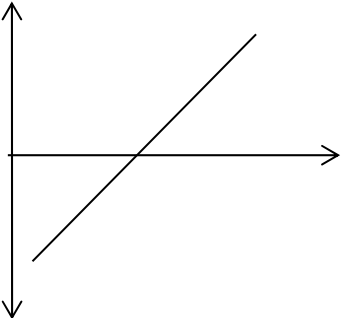


1(a)	<i>idea that</i> (cortex / pith / cells) are turgid / contain water and enlarge / contain water and expand ; <i>idea that</i> cuticle is waterproof / reduces, water loss / transpiration or epidermal cells do not expand (as much) ; <i>idea of</i> vascular tissue / xylem, provides mechanical support / strength ;	2
1(b)(i)	five stated concentrations with units ; reasonably evenly spaced ; method for proportional dilution for minimum two intermediate concentrations to produce total of 50 cm³ ;	3
1(b)(ii)	<i>independent variable:</i> sucrose concentration / molarity ; <i>dependent variable:</i> <i>idea of</i> (degree of / angle of) curvature / bending (of the strip) ;	2

Question	Answer	Marks
1(b)(iii)	any seven from: 1 strips same length / size ; 2 strips taken from same (type of) plant / stalk / species / age (of plant) ; 3 Petri dishes / containers, containing different sucrose concentrations ; 4 same / stated volume of sucrose solution / enough to cover the strips of stalk ; 5 strip(s) placed in different concentrations ; 6 cover container (to prevent evaporation) ; 7 (leave strips immersed for) same / stated, time ; 8 maintaining a constant temperature and a suitable method ; 9 record direction in which strip curved ; 10 measure / calculate (angle / degree of), curvature / bending ; 11 repeat experiment at least twice / three replicates, and finding mean ; 12 <i>ref. to</i> low risk ;	7

Question	Answer	Marks
1(c)(i)	1 x-axis sucrose <u>concentration</u> and mol dm^{-3} ; 2 y-axis degree / angle of curvature / bending ; 3 line drawn crossing x-axis once ; 4 label identifying intersection (on x-axis) as water potential equivalent to the cells ; e.g. degree angle of curvature  sucrose concentration + mol dm^{-3} or 	4
1(c)(ii)	straight or no, curvature / bending ;	1